

Learning Report

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1 Week 1

1.1 Python file processing

Task goal: create a script that takes an input .txt file, cleans it and outputs a .csv file with all words and how frequently they appear.

1.2 Dataset processing using pandas

The dataset contains information on fortune 1000 companies, including rank, revenue, profit etc. It was cleaned by removing non-numeric data e.g. CEO name, and removing symbols from numeric data so it can be analysed. The following table contains a small amount of information on dataset size and mean values. The missing values were not removed as they mostly correspond to companies that are new or were not in the fortune 1000 list the previous year.

Dataset size	Feature count	Missing value count	Mean Revenue	Mean Profit
1000 rows x 10 columns	10	402	14378.202400	1371.597015

1.3 Creating plots

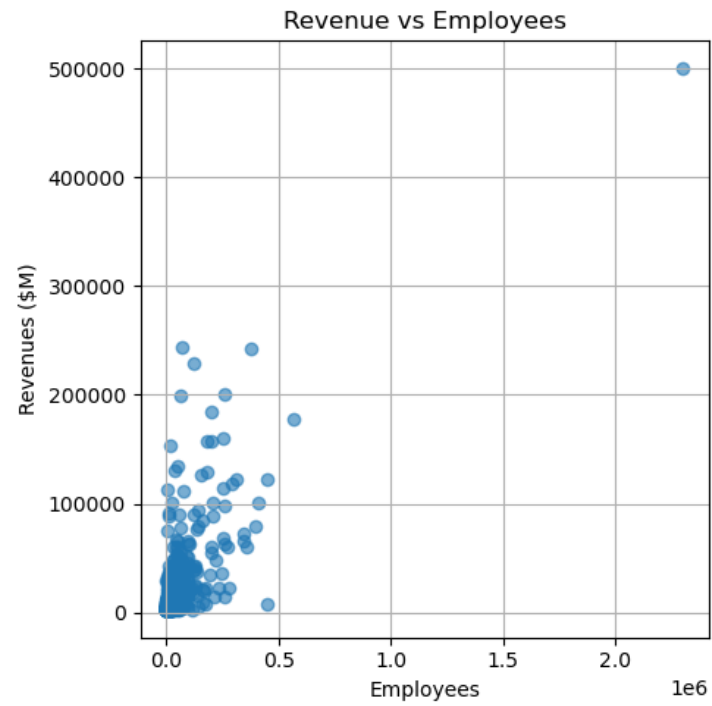


Figure 1: Revenue vs no. employees



Figure 2: Profit vs no. employees



Figure 3: Rank increase vs revenue change %

1.4 Results Analysis

Figure 1 shows a positive relationship between the number of employees and company revenue. In general, larger companies tend to generate higher revenues, although there is considerable variation among companies of similar size. A small number of very large companies appear as outliers with exceptionally high employee counts and revenues.

Figure 2 indicates that company profit does not increase as consistently with employee numbers as revenue does. While some larger companies achieve high profits, others generate relatively modest profits despite employing many people, suggesting that profitability depends on factors beyond company size alone.

Figure 3 demonstrates a clear positive relationship between revenue change and changes in Fortune 1000 ranking. Companies with higher percentage increases in revenue generally experience larger improvements in rank, whereas companies with declining revenue tend to lose ranking positions. Although the overall trend is strong, several outliers indicate that ranking can also be influenced by other business factors.

Overall, the visualisations suggest that employee count is positively associated with revenue, but is a weaker predictor of profit. Revenue growth appears to be strongly related to improvements in company ranking.

2 Week 2

3 Week 3